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ORIGINAL ARTICLE

Aerial image super-resolution based on deep recursive dense network for disaster area surveillance

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Abstract

Aerial images are often applied into disaster area surveillance. High-resolution (HR) aerial images are preferred to monitor the disaster area since they can provide abundant information. However, limited by hardware device and imaging environment, the resolution of captured aerial images may not meet the needs of practical application. Image super-resolution (SR) is an effective way to improve the resolution of captured images in a post-processing manner. Recently, convolutional neural networks (CNNs) have demonstrated great success in image SR. However, these CNN models cannot be easily applied to real-world scenarios due to requiring huge storage and computational resources. To reduce resource consumption, we need to decrease network parameters. Recursive network can effectively reduce network parameters, which motivates us to explore a more effective image SR method. In this paper, we proposed a deep recursive dense network (DRDN) to reconstruct HR aerial images. In the DRDN, the proposed recursive dense block (RDB) can fully extract abundant local features and adaptively fuse different hierarchical features of LR image for HR image reconstruction. In addition, the recursive manner of RDB in DRDN can effectively reduce the parameter of network. The experimental results on aerial images demonstrate the superiority of our proposed method.

Keywords Disaster area surveillance · Aerial image · Super resolution · Deep recursive dense network

1 Introduction

In recent years, unmanned aerial vehicles (UAVs) have been widely used in many fields, such as agriculture, military, and emergency management [1]. The UAVs are gaining outstanding success because of their versatility, flexibility, low cost, and minimized operational risk. The aerial images from UAVs can empower people to monitor dangerous areas when a natural disaster occurs [2]. In the monitoring system, high-resolution (HR) images are usually preferred because they can provide more high-frequency information compared with low-resolution (LR) images, which can improve the monitoring ability of disaster area surveillance systems. However, the aerial images from UAVs usually encounter corruption, such as dynamic blur and downsampling, which lead to the resolution of the captured aerial images may not be fit for such purpose in a monitoring system [3]. Therefore, it is valuable to enhance the resolution of already captured aerial images.

Aerial image super-resolution (SR) is a technology to recover the HR aerial image from the captured LR aerial image in a post-processing manner. Image SR is initially proposed for natural images, and some aerial image SR methods have also been proposed during recent years [3, 4]. To tackle image SR problem, numerous methods have been proposed, including interpolation-based [5], reconstruction-based [6], and learning-based methods [7–13]. Recently, deep learning (DL) acts as a very important branch of machine learning, which has achieved great success in many computer vision tasks. Dong et al. proposed the pioneer

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convolutional neural network (CNN) model for image SR. Their method, termed SRCNN [14], used a CNN to learn an end-to-end mapping between LR and HR images with little pre-/post-processing and achieved the state-of-the-art performance. The performance of SRCNN showed the superiority of DL in image SR and provides useful guidance for future research work [15]. Inspired by the success of VGG [16] on ImageNet classification that the deeper network can obtain better performance, Kim et al. increased the depth of network to 20 in VDSR [17], which achieved a significantly improved performance compared with SRCNN. However, training a very deep neural network usually faces gradient vanishing/exploding. The authors adopted residual learning and adjustable gradient clipping to solve a gradient explosion problem. Even though VDSR can obtain good performance, it requires a lot of computational costs for all convolution operations are performed in HR space. For this problem, Shi et al. proposed an efficient sub-pixel convolutional network (ESPCN) [18] for image SR, which used a sub-pixel convolution layer in the last layer to up-sample the output image into a desired resolution. Similarly, Dong et al. proposed fast super-resolution convolutional neural networks (FSRCNN) [19] for faster and better SR, which adopted a compact hourglass-shape CNN structure to accelerate the SR process. However, these two CNNs are both shallow so that they are not able to achieve satisfactory SR performance.

To ease the difficulty of training deep CNNs, He et al. proposed residual blocks to construct residual networks (ResNets) [20], which have surpassed the 1000-layer barrier. Lim et al. proposed an enhanced deep super-resolution (EDSR) [21] network by removing some unnecessary components of residual block to improve the performance of image SR and won the NTIRE2017 Super-Resolution Challenge [22]. Zhang et al. proposed a residual dense network (RDN) [23] and a residual channel attention network (RCAN) [24] for image SR, which both achieved good performance. Even though EDSR, RDN, and RCAN achieved good performance, these networks have 43M, 22M, and 16M parameters, respectively, which all need a huge amount of storage resources and are less applicable in terminal equipments.

Inspired by DenseNet [25], Tong et al. adopted a dense block to build SRDenseNet [26] for image SR, but in the SRDenseNet the growth rate is relatively small (e.g., 16), which restricts the SR performance. Tai et al. introduced memory block to construct MemNet [27]. Lai et al. proposed LapSRN [28] to progressively reconstruct the sub-band residuals of high-resolution images. As the network depth grows, the features in CNN would be hierarchical. However, these methods do not exploit the hierarchical features efficiently. As an effective way to shrink network model, recursive blocks are widely used in SR methods, such as deeply recursive convolutional network (DRCN) [29], deep recursive residual network (DRRN) [30], dual-state recurrent network (DSRN) [31], and deep recursive up-down sampling networks (DRUDN) [32]. Due to the lack of dense information flow between different layers, DRCN, DRRN, DSRN, and DRUDN do not achieve good performance. Furthermore, DRCN and DRRN interpolated the original LR image to the desired size and as the input. This pre-processing step not only increases computation complexity quadratically but also loses some details of the original LR image.

In order to address the drawbacks of the methods mentioned above, we propose a deep recursive dense network (DRDN) for aerial image SR. DRDN utilizes the original LR image as input of the whole network rather than bicubic interpolation image. Then, the proposed recursive dense block (RDB) is adopted to fully extract abundant local features and adaptively fuse different hierarchical features for HR image reconstruction. Finally, the desired size HR image is obtained by the reconstruction part in DRDN. In summary, there are mainly three contributions of this paper:

1. To effectively enhance aerial image resolution, the DRDN is proposed in this paper.
2. To achieve good HR reconstruction performance, our DRDN uses RDB to fully extract abundant local features and adaptively fuse different hierarchical features of LR image for HR image reconstruction.
3. To effectively reduce the parameter of the network, the recursive manner of RDB is adopted in DRDN.

The rest of this paper is organized as follows. In Section 2, three related works are introduced in detail. Section 3 describes the proposed method DRDN in detail. Section 4 demonstrates the experimental results. Section 5 briefly discusses the proposed DRDN. Finally, Section 6 concludes this paper.

2 Related work

Recently, DL-based methods have been widely used in-many computer vision tasks and achieved dramatic advantages compared with conventional methods [20, 33–37]. Section 1 provides an overview of DL-based image SR. This section focuses on the three most related works to our proposed method. Figure 1 uses only 6 convolutional layers to show the sim7plified network structures of these three related methods, where the activation functions (Relu) [38] and batch normalization (BN) [39] are omitted for clarity.

2.1 DRCN

DRCN [29] is motivated by the observation that adding more convolutional layers introduces more parameters,

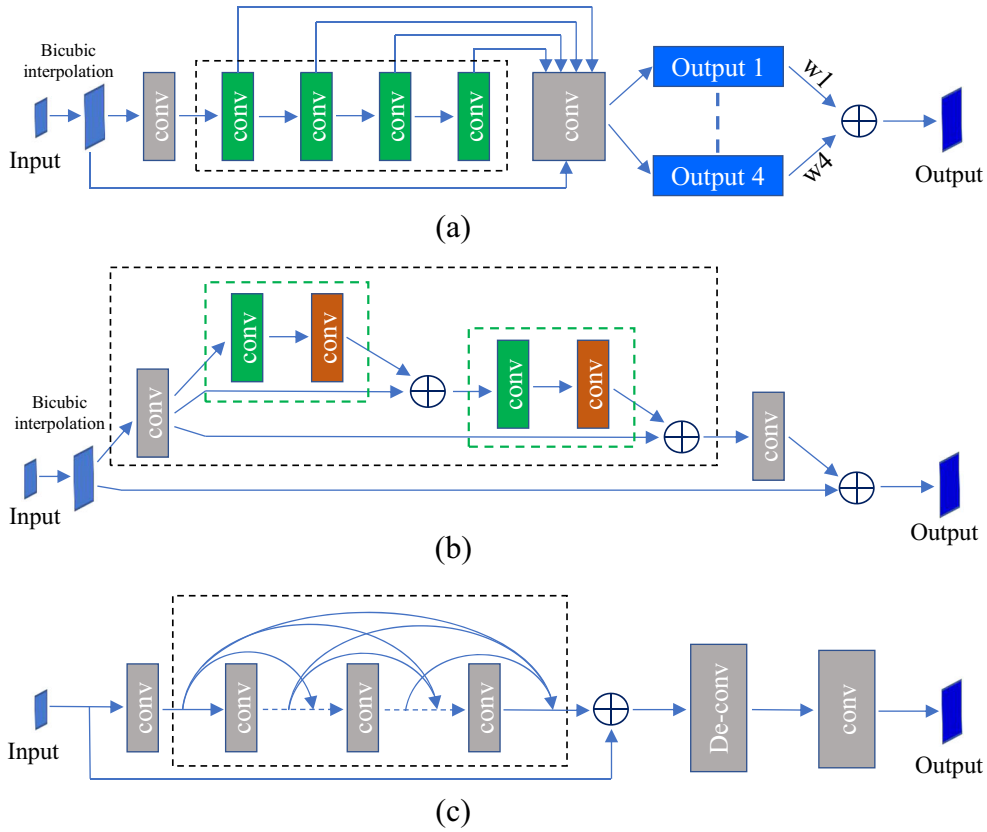


Fig. 1 Simplified network models of **a** DRCN. The black dashed box indicates a recursive layer, in which all of the convolutional layers (with dark green color) share the same parameters. **b** DRRN. The green dashed box refers to a residual unit. The black dashed box refers to a recursive block including two residual units. In the recursive block, the corresponding convolutional layers in the residual units (with green or brown color) share the same parameters. **c** SRDenseNet. The black dashed box refers to a dense block, among which all of the convolutional layers (with gray color) are densely connected, but the convolutional layers in the dense block do not share the same parameters. In all the three cases, the input and output with blue color refer to the LR image and reconstructed SR image, respectively, and ⊕ represents the element-wise addition

where the network is likely to overfit and also consumes a large amount of storage resources. DRCN introduces a recursive layer into the network so that its function is equivalent to the increase of network depth since more recursions are performed in the recursive layer while the model parameters do not increase. DRCN includes three parts: embedding net, inference net, and reconstruction net, which are shown as the left, middle 4, and right convolutional layer(s) in Fig. 1a, respectively. The embedding net ($f_1(x)$) takes the bicubic interpolation image (x) and represents it as feature maps (H_0) ready for inference. The inference net ($f_2(H_0)$) stacks T recursions in the recursive layer, which fulfills the main task of SR. The reconstruction net ($f_{\text{Rec}}(H_T)$) transforms the feature maps (H_T) back into the original image space to generate the intermediate HR image. However, since training a deep network is difficult, recursive-supervision and skip-connection are used to mitigate the training burden. Specifically, for the t-th recursion in the recursive layer, the output of the reconstruction net is formulated as:

$$y_t = f_{\text{Rec}}(f_2^t(f_1(x))) + x, \tag{1}$$

where x is the bicubic interpolation image. Each intermediate HR image y_t is learned with supervision. Finally, a weighted average strategy is adopted to obtain the HR image (y), which can be formulated as:

$$y = \sum\nolimits_t^T w_t y_t, \tag{2}$$

where weights w_t is learned during the training and T is set as 16 in [29].

2.2 DRRN

DRRN [30] used local and global residual learning to build a very deep network structure and introduced recursive learning in the residual units to keep model compact. The corresponding convolutional layers in the residual units share the same parameters. DRRN used a recursive block with a multiple paths between the input and output of the recursive block as shown in Fig. 1b. This multi-path structure can help gradient backpropagation to address the vanishing/exploding gradients problem during training. Compared with chain mode, this multi-path structure facilitates the training process and is less likely to overfitting. In DRRN, the residual unit number U in each recursive block and the recursive block number B are the key parameters. Given different U and B, we can obtain DRRN with different depths.

DRRN takes the bicubic interpolation image (x) as input. The output (y) of DRRN can be obtained by the following formula:

$$y = D(x) = f_{\text{Rec}}(R_B(R_{B-1}(...(R_1(x))...))) + x, \tag{3}$$

where f_{Rec} denotes the function of the last convolutional layer, which is used to reconstruct the residual in DRRN. R_{B-1} is the function of the B-1th recursive block.

2.3 SRDenseNet

SRDenseNet [26] introduced densely connected convolutional networks into image SR. The dense skip connections combine the low-level features and high-level features, which boost the SR reconstruction performance. In addition, the dense skip connections in the network can mitigate the vanishing-gradient problem during training very deep networks. The SRDenseNet aims to learn an end-to-end mapping between the input (LR image) and the output (HR image). It can be seen from Fig. 1c that SRDenseNet mainly contains four parts: low-level feature extraction layer, dense block, deconvolution layer, and reconstruction layer. Among these four parts, the dense block is the main component solving image SR task. In SRDenseNet, there are 8 dense blocks. The deconvolution layer is used to learn upsampling filters, which do not only contribute significantly to the whole SR reconstruction process but also improve the SR performance. The reconstruction layer outputs the HR image.

Denoting x and y to be the input (LR image) and output (HR image) of SRDenseNet. The mathematical formula of SRDenseNet can be expressed as follows:

$$y = f_{\text{Rec}}(f_{up}(D_d(D_{d-1}...(D_1(f_l(x)))))), \tag{4}$$

where f_l denotes the function of low-level feature extraction layer, and D_{d-1} denotes the function of d-1th dense block. f_{Rec} and f_{up} are the function of deconvolution layer and reconstruction layer, respectively.

3 Deep recursive dense network for aerial image SR

In this section, we introduce our proposed deep recursive dense networks (DRDN). Different from DRCN and DRRN, we design a novel recursive network. Inspired by SRDenseNet that dense connection can combine the low-level features and high-level features to improve the image SR performance, the recursive dense block (RDB) is proposed. A 1×1 convolutional layer at the last of RDB can adaptively control output information. Besides, rather than taking the bicubic interpolation image as input, we use the LR image as input of the whole network and de-convolutional layer to upsample image resolution. As shown in Fig. 2, DRDN can be divided into three parts. RDB is the main component of our network. We first introduce RDB. Then, the structure of DRDN is introduced in detail.

3.1 Recursive dense block

Here, we introduce our RDB in detail. As the core of our network, RDB block consists of groups of convolutional layers and Relu activation functions, and a 1×1 convolutional layer. Stacking RDB block for B times forms the recursive layer for parameter sharing.

For simplicity, the RDB block in Fig. 3 only has three convolutional layers to build a dense connection. In our paper, C denotes the number of convolutional layers in a RDB block. Let F_{b-1} denote the input of the b-th RDB. The output of the c-th convolutional layer of the b-th RDB can be obtained by the following formula:

$$F_{b,c} = \sigma(W_{b,c}[F_{b-1}, F_{b,1}, F_{b,2}, \cdots, F_{b,c-1}]), \tag{5}$$

where σ is the PReLU activation function and the bias term is omitted for clarity, and $W_{b,c}$ is the weight set of the c-th convolutional layer. The input (F_{b-1}) of the bth RDB and the output ($F_{b,c}$) of the c-th convolutional layer of the bth RDB all have G feature maps. $[F_{b-1}, F_{b,1}, F_{b,2}, \cdots, F_{b,c-1}]$ is the concatenation operation. It can been seen from (5) that the output of the preceding RDB has direct connections with each convolutional layer of the current RDB, which not only preserves the feed-forward features but also extracts current dense features.

The 1×1 convolutional layer can adaptively fuse the features from preceding RDB blocks and all of the convolutional layers in a current RDB block. As described above, the feature maps from the b-1th RDB block are directly connected with the b-th RDB block in a concatenation manner. This is the main difference between with dense block in SRDenseNet [26]. Let F_{b-1} be the outputs of the

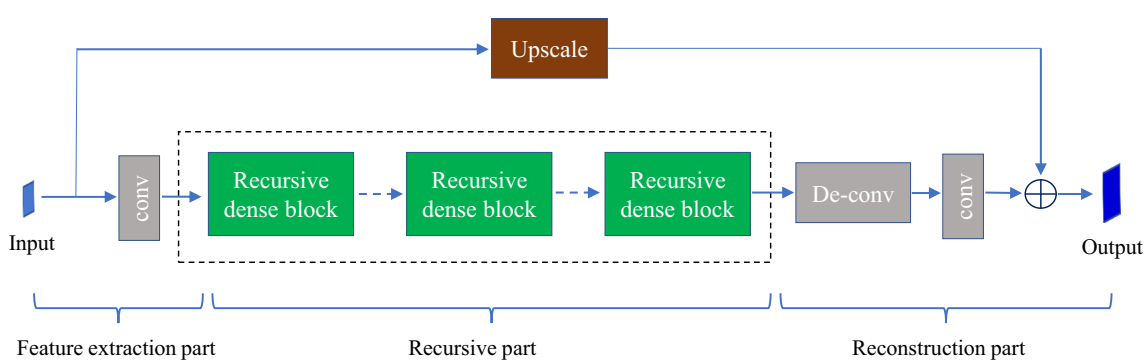


Fig. 2 The structure of deep recursive dense networks (DRDN)

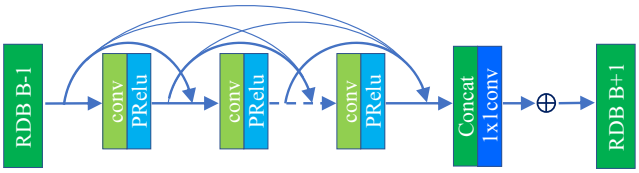


Fig. 3 The architecture of recursive dense block (RDB)

preceding RDB. $F_{b,1}, F_{b,2}, \cdots F_{b,c}$ denote the outputs of the whole convolutional layers in current RDB.The 1×1 convolution operation can be formulated as follows:

$$F_b = f_{1\times1}([F_{b-1}, F_{b,1}, F_{b,2}, \cdots, F_{b,C}]), \tag{6}$$

where $f_{1\times1}$ is the function of the 1×1 convolutional layer in the bth RDB block, and F_b is output of the bth RDB block in our DRDN. In our DRDN, all convolutional layers have G feature maps. $\left[F_{b-1}, F_{b,1}, F_{b,2}, \cdots, F_{b,C}\right]$ refers to the concatenation of the feature maps $(F_{b-1}, F_{b,1}, F_{b,2}, \cdots F_{b,c})$ which will result in $G+C\times G$ feature maps. The 1×1 convolutional layer at the last of RDB can reduce the feature maps from $G + C \times G$ to G. As a result, it can control how much of the previous features from preceding RDB should be reserved and how much of the current features in the current RDB should be stored. With the 1×1 convolutional layer, the growth rate G can be set larger. In SRDenseNet, a larger growth rate in dense block would be hard to train. It should be noted that 1×1 convolutional layer realizing feature map fusion can improve the network representation ability, which results in better SR performance.

3.2 Network structure

Figure 2 shows the structure of our DRDN, where the activation function (PRelu) following the convolutional and deconvolutional layers is omitted for clarity. It can be seen from Fig. 2 that our DRDN mainly includes three parts: feature extraction part, recursive part, and reconstruction part. Our DRDN takes as input LR images and outputs HR image. Each part of our DRRN will be described in detail in the following paragraph.

Feature extraction part Given the LR image x, a convolutional layer followed with PRelu is used to extract the low-level feature. This part can be expressed by the following formula:

$$F_l = \sigma(Conv(W_0, x)), \tag{7}$$

where F_l denotes the extracted low-level feature. σ and $Conv\,(\bullet)$ refer to the PRelu activation function and the convolution operation, respectively. In the convolution operation, a filter with 3×3 kernel size and stride of 1 are employed [16]. W_0 represent the corresponding weight. F_l is not only the output of low-level feature extraction part, but also the input of the following recursive part.

Recursive part This part plays the most important role in DRDN, which mainly accomplishes the task of recovering high-frequency information. In Fig. 2, the black dashed box refers to the recursive dense block (RDB), among which all of the dense block (which green color) share the same parameters. It accepts the low-level feature from previous part and recursively learns the mapping relationship between LR image and HR image. Our proposed recursive dense block can make the most use of hierarchical features of LR image and effectively reduce the parameter amount of the whole network. The recursive part consists of B recursive dense blocks. All of the B recursive dense blocks share the same parameters. The entire recursive part can be summarized by the following formula:

$$F_B = f^B(F_l), \tag{8}$$

where f^B denotes the function of recursive part with B recursive dense blocks. F_B is the output of the recursive part.

Reconstruction part We assume that the feature map from the previous part is the size of $C_f \times H \times W$, where C_f denotes the number of feature map channel. The H and W of feature maps are equal to the height and width of the LR image since the proper padding strategy is used. The size of number features will not change in the convolution operation of previous parts. The reconstruction part needs to transform the feature maps back into the HR image space.

In our DRDN, the reconstruction part mainly includes two branches: coarse branch and residual branch. In the coarse branch, the original LR image is upscaled to the desired size. The upscaled coarse can be obtained by

$$F_{coarse} = f_{UP}(x), \tag{9}$$

where f_{UP} represents the operation of an upsample kernel. In our DRDN, we select a bilinear upsample kernel.

In the residual branch, the residual feature F_B is first upscaled by a deconvolutional layer. Then, a 3×3 convolutional layer uses the upscaled feature to recover a residual image. In mathematical calculation, the deconvolution can be seen as an inverse operation of the normal convolution. In normal convolution operation, the image is convolved by a filter with a stride k, and the output image size is $1/k$ times of the input image. Contrarily, in deconvolution operations, the size of output image will be k times of the input image. Through the deconvolution layer, the size of feature maps is upscaled to the desired size of HR image, but the number of feature maps channel is not desired. The 3×3 convolutional layer transforms the feature maps back into the original image space.

In our DRDN, the residual image can be obtained by the following formula:

$$F_{residual} = f_{residual}(F_B), \tag{10}$$

where $F_{residual}$ denotes the reconstructed residual image, and the $f_{residual}$ is the function of the residual branch. F_B is obtained by (8).

Finally, an element-wise addition strategy is adopted to obtain the HR image (y), which can be represented as

$$y = F_{residual} + F_{coarse}. \tag{11}$$

Given the LR image x, the goal of our DRDN is to recover a SR image, which is similar to the ground truth image (HR image). The L_1 loss function is selected to train our DRDN. There are two advantages to choose L_1 loss compared with L_2 loss [40]. First, the reconstructed image obtained by using L_1 loss is more suitable for human vision system, which can preserve more abundant textures and local structures compared with L_2 loss. Second, L_1 loss is easier to find a better minimum solution compared with L_2 loss.

So, for N training samples, the L_1 loss function can be expressed by the following formula:

$$L(\Theta) = \frac{1}{N}\sum_{i=1}^{N}\left\| HR^i - SR^i \right\|_1, \tag{12}$$

where Θ is the parameter set of the whole network.

4 Experimental results

4.1 Implementation details

In the design of network structure, PReLU [33] is used as the activation function following all of the convolutional and deconvolutional layers except the reconstruction part. For all convolutional layers, the kernel size is set as 3×3, the stride 1, and the padding 1. But for deconvolutional layers, by reference to [41], we set various k, s, and p in $Deconv(k, s, p)$ to perform upsampling operations for different scale factors. k, s, and p represent the kernel size, stride, and padding, respectively. For ×2 scale factor, the k, s, and p in $Deconv(k, s, p)$ are set to 6, 2, and 2, respectively. For ×3 scale factor, the k, s, and p in $Deconv(k, s, p)$ are set to 7, 3, and 2, respectively. For ×4 scale factor, the k, s, and p in $Deconv(k, s, p)$ are respectively set to 8, 4, and 2.

In our experiments, we choose the DIV2K [42] dataset as the training dataset, which contains 800 high-quality training images. Because of the richness of this dataset, many recent image SR methods [21, 23, 24, 32, 41] use DIV2K as well. To obtain LR DIV2K training images, we employ bicubic interpolation in MATLAB2017a to downscale HR images with the scaling factors (2, 3, and 4). Following the settings of [21], the minibatch size and LR size are set to 16 and 48 for DRDN, respectively. We randomly augment the patches by flipping horizontally or vertically and rotating 90°. One thousand iterations of back-propagation constitute an epoch. In the training stage, Adam [43] is used to optimize DRDN. The initial learning rate of Adam is set as 0.0002, and it is halved every 200 epochs. The number of convolutional kernels in the last layer and the other layers is set to G_{out} and G, respectively. Because our DRDN mainly focuses on the reconstruction of RGB image, G_{out} is set to 3. In the testing stage, as our focus is aerial image SR, the standard benchmark datasets of natural image are not suitable for testing. Therefore, we create a test dataset containing 18 HR aerial images, which all are from Flickr using keywords such as aerial images of disaster areas, earthquake, flood, and tornado. As shown in Fig. 4, the test dataset covers different types of disaster areas. The SR reconstruction results are evaluated objectively with PSNR and SSIM [44]. A higher value of PSNR or SSIM

Fig. 4 The test dataset of aerial images

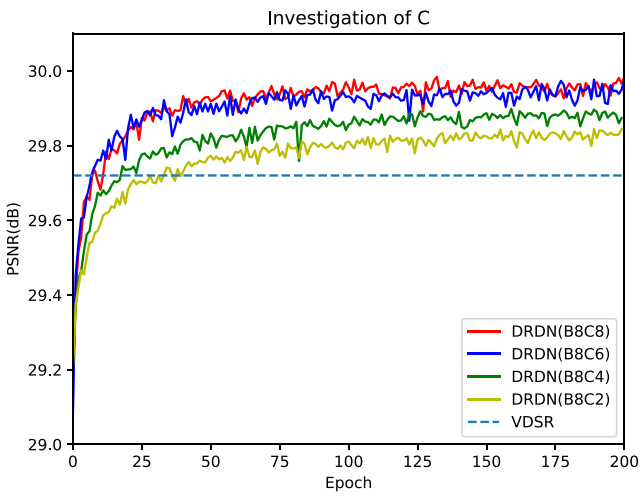


Fig. 5 Convergence analysis of DRDN with different values of C

means a better result. To keep consistency with previous methods, the reconstruction results of different methods are only evaluated on luminance (Y) channel of YCbCr space. All experiments of DRRN are conducted based on Pytorch 1.1.0 on a desktop with an NVIDIA 1080 Ti GPU.

4.2 Study of the number of convolutional layers per RDB and the number of RDB

In this section, we explore the basic parameters of DRDN: the number of convolutional layers per RDB (denote as C for short) and the number of RDB (denote as B). In the following experiments, the number of convolutional kernels G is set to 32. Each model is trained under 200 epochs and evaluated on the dataset of aerial images with scale factor $\times 4$.

At first, to validate the influence of the parameter of C, B is set to 8. Experiments are conducted on four combinations, namely B8C2, B8C4, B8C6, and B8C8. The PSNR visualization of convergence is shown in Fig. 5. We can see that the performance of our DRDN can get a higher performance with increasing C. But, the difference of B8C6 and B8C8 is very small. This is probably because our RDB in DRDN has strong representational ability and does not need more convolutional layers.

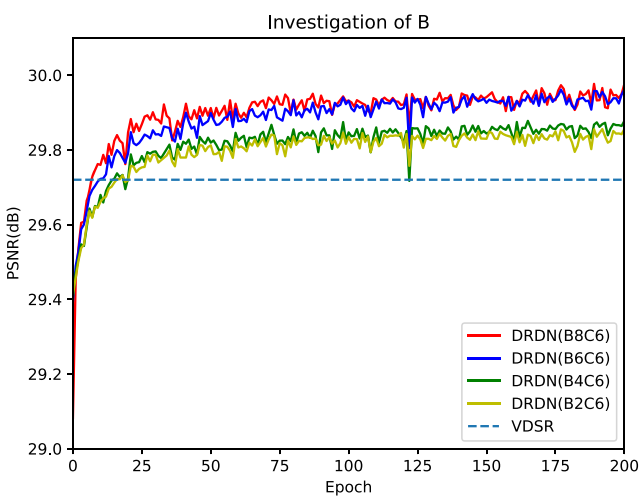


Fig. 6 Convergence analysis of DRDN with different values of B

Secondly, to study the influence of B, we set C to 6. We test the performance of these models: B2C6, B4C6, B6C6, and B8C6. It can be seen from Fig. 6 that larger B can lead to higher performance. However, there is little difference between B6C6 and B8C6 in performance.

4.3 Comparison with the state-of-the-arts

In this section, the quantitative and qualitative comparisons are provided in detail. We compare our DRDN with seven state-of-the-art methods: SRCNN [14], VDSR [17], DRCN [29], LapSRN [28], SRDenseNet [26], SRMDNF [34], and DRUDN [32]. All results of comparison methods are obtained using the public code provided by the corresponding author. According to the section of 4.2, in comparison with the state-of-the-arts, both B and C are set to 6. As most methods [28, 32] for comparison use 64 filters per convolutional layer, the number of convolutional kernels G is set to 64 for fair comparison.

Table 1 lists the average PSNR/SSIM for scale factors $\times 2$, $\times 3$, and $\times 4$ of different methods on aerial image dataset. As can be seen from Table 1, the proposed DRDN achieved the best performance on both PSNR and SSIM metrics among these comparison methods. The qualitative comparison of DRDN with other different methods is shown in Fig. 7, which indicates our DRDN can reconstruct a faithful SR image with clearer texture and details compared with other methods. Therefore, our proposed method demonstrates the superiority compared with state-of-the-arts both in quantitatively and qualitatively.

4.4 Analysis of network parameters

The number of parameters plays an important role in determining the potential of CNN networks. More parameters usually need more images for training to avoid overfitting and can obtain better performance. However, it needs more storage and computational resources. Our aim here is to obtain better performance with fewer parameters. In Fig. 8, we illustrate the parameters-to-PSNR relationship of our model and seven state-of-the-art methods, including SRCNN [14], VDSR [17], DRCN [29], LapSRN [28], SRDenseNet [26], SRMDNF [34], and DRUDN [32].

It can be seen from Fig. 8 that the proposed method DRDN benefits from the recursive manner of RDB and obtains higher parameter efficiency compared with other

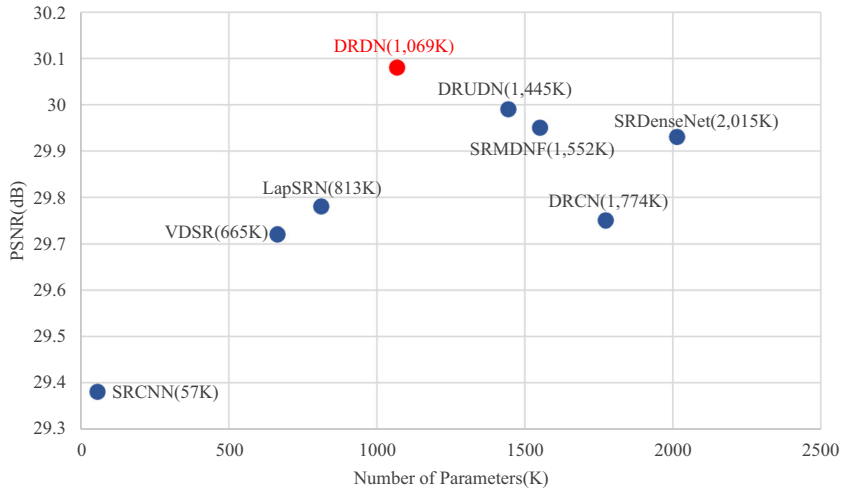
Table 1 Quantitative comparison for scale factors ×2, ×3, and ×4 of our DRDN with other different methods on the dataset of aerial images

Methods	×2		×3		×4	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Bicubic	32.66	0.8837	29.91	0.8106	28.46	0.7593
SRCNN [14]	34.44	0.9134	31.12	0.8424	29.38	0.7881
VDSR [17]	34.84	0.9183	31.51	0.8516	29.72	0.8001
DRCN [29]	34.92	0.9172	31.58	0.8512	29.75	0.7994
LapSRN [28]	34.86	0.9181	-/-	-/-	29.78	0.8010
SRDenseNet [26]	35.10	0.9196	31.70	0.8532	29.93	0.8038
SRMDNF [34]	35.11	0.9199	31.69	0.8533	29.95	0.8043
DRDUDN [32]	35.05	0.9194	31.70	0.8533	29.99	0.8050
DRDN(ours)	**35.27**	**0.9210**	**31.83**	**0.8550**	**30.08**	**0.8068**

The best performance is shown in bold

Fig. 7 Qualitative comparison of DRDN with other different methods on ×4

Fig. 8 The number of parameters of different SR methods.The PSNR values are evaluated on the dataset of aerial images with scale factor ×4. Red point denotes our method



state-of-the-art methods. This further verified the superiority of our proposed method.

5 Discussions

When the proposed DRDN has unfolded, the longest depth has 45 weight layers. The SR task can benefit from our DRDN in three aspects: (a) the dense connection in RDB block provides an effective way to reuse feature maps that are already learned and forces the current layer to learn complementary information, thus avoiding the learning of redundant features; (b) due to the use of 1×1 convolutional layer in RDB, different hierarchical features of LR images can be adaptively fused for HR image reconstruction; (c) because of the recursive manner of RDB, our DRDN can obtain higher parameter efficiency compared with other methods. So, the proposed method DRDN can obtain a better performance with fewer parameters.

However, the proposed method should be further improved in terms of the time consumption. Moreover, our DRDN only takes the super-resolution of several integer scale factors ($\times2$, $\times3$, $\times4$) into consideration.

6 Conclusions

In this paper, we proposed a DRDN for aerial image super-resolution. In the DRDN, we proposed an effective RDB to improve the representational capability of the network. The RDB can fully extract abundant local features and adaptively fuse different hierarchical features of low-resolution (LR) image for the high-resolution (HR) image reconstruction. In addition, the recursion manner of RDB in DRDN can effectively reduce the parameter amount of networks, which helps reduce the consumption of computational resources and storage resources. The experimental results on aerial images verify the superiority of our proposed method in both subjective effects and objective evaluation.

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